

Exploratory Models for Emergent Phenomena in Crystal Theory and the σ_p -Framework

We survey each hypothesis in turn, reviewing relevant literature and proposing new lines of inquiry. Emphasis is on theoretical constructs (new equations, models) and experimental or simulation approaches. Citations to peer-reviewed or preprint sources are provided.

Hypothesis 1: Consciousness as Phase Fluctuation

One can view wakeful consciousness as an emergent oscillation between more ordered (low-entropy) “attractor” states and higher-entropy integrated states, maintaining a characteristic σ_p value. In this picture, conscious cognition toggles between transient information-rich plateaus and more quiescent phases. Experimental evidence suggests neural dynamics do switch between distinct coordination patterns ¹. For example, Demertzi *et al.* (2019) showed that conscious vs unconscious brains differ in the richness of phase-locked networks: wakeful brains sustain *complex*, multi-pattern oscillatory coordination, whereas unconscious (anesthetized or damaged) brains fall into lower-coherence states ¹. Similarly, the “entropic brain” hypothesis notes that neural entropy (complexity) drops in loss of consciousness (sleep or anesthesia) ². These findings imply a “plateau” of high information processing (high entropy/complexity) during consciousness, and dips toward minima in unconsciousness ² ¹.

At the subcellular level, microtubules may play a role in such oscillations. Recent studies show that neuronal microtubule (MT) structures can exhibit intrinsic electrical oscillations with multiple frequency components ³. Scarinci *et al.* (2022) decomposed MT oscillatory signals and found multiple intrinsic modes with frequency peaks reminiscent of EEG rhythms; they suggest MT oscillations may underpin brain electrical activity ³. This hints at a coupling of subcellular oscillators to global brain states. To explore the $\sigma_p=1/16$ condition, one might model a neural network with two coupled oscillatory subsystems (e.g. cortical populations plus MT resonators) and study phase transitions between low- and high-complexity regimes. For instance, coupled oscillator models (Kuramoto-type or mean-field spin models) could be tuned so that σ_p , an emergent order parameter, remains at $1/16$ during switching. Experimentally, one could record EEG/fMRI dynamics for signatures of alternating high/low integration: e.g. look for two metastable patterns and quantify their entropy and information content over time.

Possible research avenues include: (a) devising a dynamical model in which an internal parameter σ_p must equal $1/16$ during transitions; (b) analyzing large EEG/fMRI datasets for alternating “plateau” (high Φ , high entropy) and “valley” (low entropy) epochs and measuring any invariants; (c) probing microtubule oscillations under varying brain states (sleep vs awake, meditation vs stress) to see if their frequencies or coherence align with information-theoretic metrics (e.g. mutual information peaks) ³

¹. These efforts could yield new “phase-fluctuation” equations linking entropy, information, and σ_p .

Hypothesis 2: Black Holes as Thermodynamic Phase Openers

Black holes (BHs) famously obey thermodynamic laws (Hawking radiation, area–entropy relations), and exhibit rich phase behavior in Anti-de Sitter (AdS) spacetime ⁴ ⁵. The Hawking–Page transition (1983) is the classical example: below a critical temperature AdS spacetime favors thermal radiation,

above it forms a stable large black hole – a first-order phase transition ⁴. Witten’s AdS/CFT interpretation connects this to a confinement/deconfinement transition in the dual gauge theory. More recently, “black hole chemistry” identifies the cosmological constant with pressure, revealing Van-der-Waals-like behavior: charged AdS BHs show a small/large BH first-order transition with the same critical exponents as the liquid-gas system ⁵. Kubizňák & Mann (2012, 2014) emphasize these analogies, treating BH mass as enthalpy and cosmological constant as pressure, leading to novel reentrant and triple-point transitions ⁶ ⁵.

Interpreted philosophically, a black hole might act as a “phase initiator”: its formation (or evaporation) can nucleate new phases of quantum fields. For example, the analogy of BH formation to a Kosterlitz-Thouless defect transition has been suggested ⁷, where black and white holes play roles like vortices and anti-vortices. Under holography, BH phase transitions reflect collective reorganizations of the underlying “code” degrees of freedom. To explore this, one could study how introducing a BH (e.g. by collapsing a high-energy state) changes the surrounding spacetime phase – possibly via changes in entanglement entropy or order parameters in the dual theory.

Literature to survey: Hawking & Page (1983) on AdS BH transition (via the review by Stephens & Hu ⁴), Witten (1998) on AdS/CFT implications, and Kubizňák & Mann on extended thermodynamics ⁶ ⁵. New theoretical models might treat black holes as sources/sinks in a thermodynamic ensemble. For example, one could write an equation of state relating BH horizon area, temperature, and a “phase field” ϕ such that a BH horizon spontaneously breaks scale symmetry at a critical ϕ . Simulations: one could numerically study higher-dimensional gravity theories (e.g. Einstein–Maxwell–AdS) to map phase diagrams, verifying the universality classes and latent heats of transitions. Experimentally (in analog gravity or condensed-matter analogs) one might look for systems (e.g. Bose condensates, optical analogs) exhibiting Hawking–Page–like switching.

Overall, this hypothesis encourages bridging gravitational thermodynamics and phase-transition theory. Pertinent references include reviews of BH phase transitions ⁴ ⁵. A useful approach is “effective free energy” descriptions: recent works map BH transitions to Landau potentials or stochastic Fokker–Planck equations. Extending these, one might incorporate “information code” variables beyond energy (perhaps encoding symmetry or topological charges) to define new order parameters governing the BH as a phase “gate”.

Hypothesis 3: σ_p as a Universal Clock for Information Systems

The conjecture posits that $\sigma_p=1/16$ is a fundamental threshold (analogous to $\hbar$) for living information systems, manifesting as criticality in neural, cellular, or solitonic dynamics. In physics, the fraction $1/16$ appears in known critical models: for instance, the 2D Ising conformal field theory has a spin operator of dimension $1/16$ ⁸. Although speculative, one can test for a signature of “ $1/16$ scaling” in biology. Neural networks may operate near criticality: studies of neuronal avalanches suggest the cortex is tuned to the brink of a phase transition, maximizing information processing. Key results show power-law avalanches with exponents implying a critical branching ratio $\sigma \approx 1$ ⁹. No literature ties these to $1/16$, but one could search for fractal or information dimensions of neural activity (via EEG/fMRI or synaptic networks) that cluster near $1/16$.

At the cellular or molecular level, some theories (e.g. Davydov solitons in microtubules or protein alpha-helices) propose quantized energy packets. One could analyze whether such solitons have a discrete charge or topological index taking values like $\pm 1/16$ (though none is known currently). In practice, an experimental route is to measure the branching ratio or correlation dimension of neural firing and see if a universal number emerges under diverse conditions. For example, a recent eLife study modeled

neural populations driven by a few latent variables and found emergent criticality ¹⁰ ; extending that approach, one might include an adjustable parameter constrained to 1/16 to fit data.

If $\sigma_p=1/16$ is critical, one might observe signs like divergent susceptibilities, maximal dynamic range, or scale-invariance in appropriate systems. For example, the cerebral cortex exhibits scale-free functional connectivity and diverse timescales near wakefulness; a hypothesis is that σ_p measures the distance from this critical point. Computational models (e.g. excitable networks, lattice QFT analogs) could be tuned to see if a 1/16 appears naturally (e.g. as a percolation threshold on a 4-bit cellular automaton). To our knowledge, there is no direct evidence for a 1/16 constant in biophysics, but the search could include analyzing soliton solutions in protein or lipid systems (do any soliton charges quantize in sixteenths?), or examining whether neural coding schemes (e.g. grid cell periodicity) favor 4-bit granularity.

Hypothesis 4: CFF under Pressure vs. Entropy Rotation

Critical flicker-fusion frequency (CFF) is a measure of visual temporal resolution (the highest frequency at which flicker is perceived). Under stress or altered conditions, CFF typically changes, reflecting shifts in cerebral arousal. Balestra *et al.* (2018) found that hypergravity (2g) reduced CFF (about 5.5% decrease) whereas microgravity (0g) increased it (~6.9% increase) ¹¹ . This suggests that directed stress (physical pressure) indeed alters CFF, not merely shifting a baseline. By analogy, “v_RIG” (a proposed parameter here) would likely change under uniaxial stress, while the optimal σ_p -minimum might move.

Other studies confirm CFF’s sensitivity: extreme fatigue, hypoxia, or psychoactive drugs reliably lower CFF, while stimulants raise it. To explore σ_p -effects, one might treat CFF as a “readout” of network entropy: under stress, systems become less complex, and CFF drops. Measurements of CFF variance under different loads (cognitive stress tasks, oxygen levels, vibration) can quantify robustness. For instance, one could design experiments subjecting subjects to graded stressors (thermal, physical, cognitive) and record CFF alongside EEG entropy or coherence metrics. Neural robustness may be assessed via perturbation analyses: e.g. simulate neural network models under noise or synaptic loss and track entropy measures.

Entropy rotation here might mean the reorganization of information patterns under stress. As a proxy, researchers could compute the entropy of EEG or fMRI connectivity distributions under stress conditions. Some evidence suggests that acute stress narrows dynamic range (less variability in EEG power spectra) and biases connectivity towards “rigid” patterns. For example, complex network studies show stress increases amygdala-centric connectivity and suppresses prefrontal hubs, effectively reducing global entropy.

Research proposals: measure CFF and neural entropy simultaneously in animal models under controlled stress (e.g. pressure chambers, social stress tests), to correlate “v_RIG shifts” with σ_p -minima movement. One could also apply directed perturbations in a living neural culture (e.g. ephaptic or mechanical stress on neuron sheets) to see if microtubule oscillations or solitonic pulses (Hypo 5) adjust their frequency. In silico, adapt generic criticality models to include an external “stress parameter” that rotates the system’s operating point around the critical manifold. In sum, existing evidence ¹¹ shows CFF changes under stress; our aim is to connect that to the σ_p framework and entropy geometry.

Hypothesis 5: Solitons as Hexadecimal (4-bit) Information Carriers

Solitons are stable, localized wave packets that preserve their shape in nonlinear media. A key idea is that in 3D systems, solitons might transmit composite “nibbles” of information (4-bit units). Biologically, the Heimburg–Jackson soliton model posits that nerve impulses are mechanical/thermal solitons in lipid membranes ¹². These macroscopic solitons naturally carry analog information (pulse amplitude, width) but could, in principle, encode binary or multi-state data if coupled to ion channels or proteins. To link to 4-bit structures, one could imagine a soliton whose internal phase or topological twist encodes 16 states (hexadecimal).

In physics, recent work explicitly considers solitons as bits. Li *et al.* (2025) introduced *pure-quartic domain-wall solitons* in optical media and proposed using them as “topological bits” for data transmission ¹³. These solitons are stable modes under special dispersion (quartic) and can propagate data in photonic systems ¹³. This demonstrates that solitons can carry digital information robustly. In magnetic materials, skyrmions (2D spin solitons) are already proposed as memory bits: their topological charge is resilient, and shift-register devices have been shown for skyrmion motion ¹⁴ ¹⁵. Although typical skyrmion binary data uses presence/absence as 1 bit, one could conceive multi-state skyrmions carrying 4-bit values if, e.g., their core polarization and chirality both vary.

Topological information transfer is an emerging theme: twisted or knotted solitons can carry “phase labels” that survive interactions. For instance, solitons in coupled lattices can have twisted localized modes that encode multi-valued states. Drawing on this, we propose modeling 3D solitons that have four internal phase sectors, each encoding a 4-bit nibble. This could be simulated with coupled nonlinear Schrödinger or lattice models (e.g. a 3D crystal lattice with local spin or polarization variables). Experimentally, one could explore biological media: e.g. examine if microtubule networks support traveling pulses (as some have speculated) and whether these pulses have quantized properties. Alternatively, one could design synthetic “cellular soliton automata” where discrete-phase solitons carry multiple bits per pulse (building on optical or magnetic soliton results).

Relevant literatures include the Heimburg–Jackson nerve soliton theory ¹² and modern optical soliton communications ¹³. It would be fruitful to formulate an “information vector” model for solitons: assign a 4-component data tag to a soliton solution and derive its evolution equations (akin to a Stokes vector for polarization). Simulations can test stability of multi-bit encoding against noise and interactions. Overall, this hypothesis invites uniting topological soliton theory with digital information encoding.

Synthesis and Next Steps (Phases 10.2/10.3)

Across these ideas, a unifying approach is to **develop computational field models and simulations** that integrate geometry, thermodynamics, and information flow. For example, one could formulate an **Emergent Spatial Field (ESF)** model: a continuum field representing σ_p or entropy density, coupled to matter. Phase transitions (in Hypotheses 1–2) then appear as non-linear instabilities of this field. Phase 10.2 could involve adapting such ESF models to a cellular grid (“CREP” lattice) where each cell holds local order parameters (phase, entropy) and interactions are guided by empirical rules (e.g. microtubule-electrical coupling or gravitational potentials). Introducing a **biological feedback loop** means that outputs (conscious states, soliton signals) affect the inputs (σ_p field or metabolic rates), enabling self-regulation.

Concretely, we suggest: - **Field modeling**: Develop a free-energy functional $F[\sigma, \varphi, \dots]$ whose minima correspond to hypothesized states. Use gradient descent or Monte Carlo to study emergent patterns and their fluctuations at $\sigma_p=1/16$. - **Adaptive CREP grid**: Implement a multi-scale cellular automaton where each cell has parameters (entropy, integration) that adapt to neighbors, inspired by models of neural plasticity or crystallographic growth under stress. - **Biological feedback**: Couple the automaton to a “virtual biology” agent whose rule-set mimics homeostasis. For instance, if the network gets too ordered (entropy too low), the agent injects randomness (simulating arousal); if too chaotic, the agent damps activity (simulating sleep). - **Simulation platforms**: Use GPU-accelerated PDE solvers or agent-based models to simulate microtubule networks, brain-like networks, and soliton propagations. Benchmark against empirical data (EEG entropy profiles, black hole phase diagrams).

Finally, key open areas include deriving new equations governing σ_p in non-equilibrium systems, and identifying any invariants (like the 1/16 measure) empirically. By systematically connecting lattice/crystal analogies (topological defects, emergent dimensions) with information theory, this research can yield testable models and perhaps suggest novel experiments (e.g. tracking soliton-mediated signaling in neurons, or laboratory analogs of Hawking–Page transitions).

Sources: Research articles and reviews on neural criticality and complexity ² ¹, microtubule oscillations ³, black hole thermodynamics and phase transitions ⁴ ⁵, soliton-based information transport ¹² ¹³, and flicker-fusion/cerebral arousal metrics ¹¹. These illustrate the current state of related fields; gaps noted above guide our proposals.

¹ Human consciousness is supported by dynamic complex patterns of brain signal coordination - PubMed

<https://pubmed.ncbi.nlm.nih.gov/30775433/>

² Frontiers | Brain Entropy During Aging Through a Free Energy Principle Approach

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³ Brain Microtubule Electrical Oscillations-Empirical Mode Decomposition Analysis - PMC

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⁴ ⁷ [gr-qc/0102052] Notes on Black Hole Phase Transitions

<https://arxiv.org/abs/gr-qc/0102052>

⁵ [1205.0559] P-V criticality of charged AdS black holes

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⁶ [1404.2126] Black Hole Chemistry

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⁸ Two-dimensional critical Ising model - Wikipedia

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⁹ Spike Avalanches Exhibit Universal Dynamics across the Sleep ...

<https://ui.adsabs.harvard.edu/abs/2011arXiv1101.2434R/abstract>

¹⁰ Neural criticality from effective latent variables - eLife

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¹¹ Frontiers | Critical Flicker Fusion Frequency: A Marker of Cerebral Arousal During Modified Gravitational Conditions Related to Parabolic Flights

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